

Soft Max Classifier [Multi Class Classification]:

$$\text{Softmax} = \hat{y}_k = \frac{e^{z_k}}{\sum_j e^{z_j}}$$

$$\text{Cross Entropy Loss} = L = - \sum_k y_k \log(\hat{y}_k)$$

$$\text{Gradient Descent} = \frac{dL}{dz_i}$$

$$= \frac{d}{dz_i} \left[- \sum_k y_k \log \left(\frac{e^{z_k}}{\sum_j e^{z_j}} \right) \right]$$

$$= \frac{d}{dz_i} \left[- \sum_k y_k (z_k - \log \sum_j e^{z_j}) \right] \because \frac{dz_k}{dz_i} = \begin{cases} 1 & k=i \\ 0 & k \neq i \end{cases}$$

$$= - \sum_k y_k \frac{d}{dz_i} (z_k - \log \sum_j e^{z_j})$$

$$= (y_i - \frac{d}{dz_i} (\log \sum_j e^{z_j}))$$

$$= \left[y_i - \frac{e^{z_i}}{\sum_j e^{z_j}} \right] = -(y_i - \hat{y}_i) \Rightarrow \frac{dL}{dz_i} = \hat{y}_i - y_i$$

MLP (Back Propagation):

$z = XW + b \equiv$
$L(z) \equiv \text{Loss}$
$\frac{\partial L}{\partial W} \equiv \text{Gradient}$

$$z_j = \sum_k X_k W_{kj}$$

$$\frac{\partial L}{\partial X} \equiv \text{Error passed to prev. layer.}$$

$$\frac{\partial L}{\partial z} \equiv \text{Error at layer output.}$$

$$\frac{\partial L}{\partial W} = \frac{\partial L}{\partial z} \times \frac{\partial z}{\partial W} \because \left[z_j = \sum_k X_k W_{kj} \right]$$

$$\therefore \left[\frac{\partial z_j}{\partial W_{ij}} = X_i \right]$$

$$\text{Let } \left[\frac{\partial L}{\partial z} = \delta_j \right]$$

$$\Rightarrow \frac{\partial L}{\partial W} = \frac{\partial L}{\partial z} \cdot X_i$$

$$\Rightarrow \left[\frac{\partial L}{\partial W} = X_i \delta_j \right] = X^T \delta$$

$$\therefore \left[\frac{\partial z}{\partial X} = \frac{\partial}{\partial X} \left(\sum_k X_k W_{kj} \right) = W_{ij} \right]$$

$$\frac{\partial L}{\partial X} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial X} \quad \text{Backprop to prev. Layer}$$

$$= \delta_j \cdot W^T$$

$$\frac{\partial L}{\partial W} = X^T \delta \quad \text{Matrix Form}$$

$$\delta_{\text{prev}} = (\delta W^T) \times f'(z)$$

Or

$$\delta^{(l)} = [\delta^{(l+1)} W^{(l+1)T}] \cdot f'(z^{(l)})$$

$\delta^{(l+1)} \equiv$ Error at next layer
$W^{(l+1)} \equiv$ Weights connecting curr. layer to next
$W^{(l+1)T} \equiv$ Reversed Wts

$$f'(z^{(l)}) \equiv \text{Derivative of activation fn at this layer.}$$

PCA (Principal Component Analysis) :

$$X \in \mathbb{R}^{n \times k} \quad n \text{ samples, } k \text{ features}$$

$$I \quad \mu = \frac{1}{n} \sum_{i=1}^n x_i \quad X_{\text{centred}} = X - \mu$$

$x^T v \equiv \text{Proj}^n$ of data point x onto v .

$$\text{Var}(v) = \frac{1}{n} \sum_{i=1}^n (x_i^T v)^2$$

$$\Rightarrow \text{Var}(v) = v^T \frac{1}{n} (X_c^T X_c) v$$

$$\text{Let } \Sigma = \frac{1}{n} X_c^T X_c \quad [\text{Covariance Matrix}]$$

$$\Rightarrow \text{Var}(v) = v^T \Sigma v \quad \text{Constraint} \Rightarrow \Sigma v = \lambda v$$

$$\text{OPTIMIZATION} \Rightarrow v^T \Sigma v = \lambda v^T v$$

$$\text{PROBLEM} \Rightarrow v^T \Sigma v = \lambda (v^T v)$$

$$\Rightarrow \boxed{v^T \Sigma v = \lambda} \quad \text{Variance along direc}^n v \text{ equals its eigenvalue.}$$

$$\boxed{\begin{array}{l} \text{Unit Direc}^n \text{ w max}^n \\ \text{Vars} \\ \max_v (v^T \Sigma v) \text{ w } v^T v = 1 \end{array}}$$

$$L(v, \lambda) = v^T \Sigma v - \lambda (v^T v - 1) \quad \left[\begin{array}{l} \text{Using Lagrangian} \\ L(v, \lambda) = f(v) - \lambda(g(v)) \end{array} \right]$$

$$\frac{\partial L}{\partial v} = 2 \Sigma v - 2 \lambda v$$

$$\Rightarrow 2 \Sigma v = 2 \lambda v$$

$$\Rightarrow \boxed{\Sigma v = \lambda v}$$